

PROJECT DESIGN PHASE

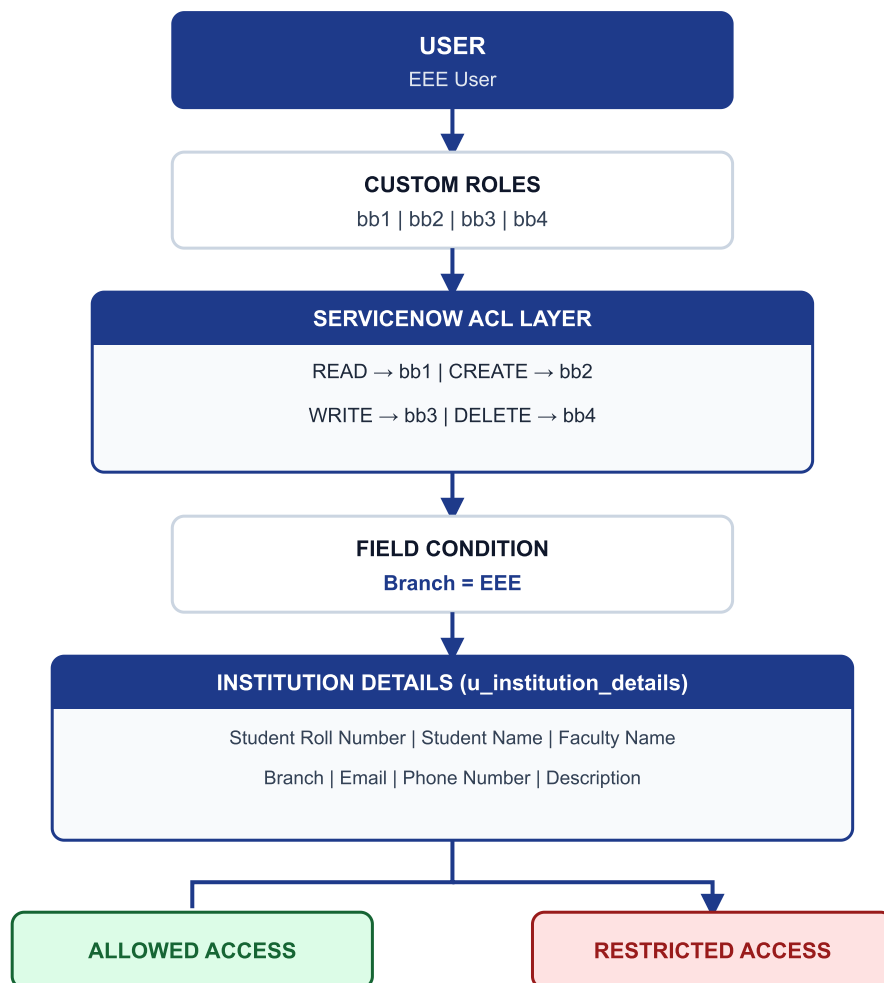
SOLUTION ARCHITECTURE

Date	3rd October 2026
Team ID	6ab8c0aa8c7c66e70673078c
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	

4.3 SOLUTION ARCHITECTURE

The solution architecture defines how the users, roles, ACLs, field conditions, and Institution Details table interact within the ServiceNow environment.

High-Level Architecture



Architecture Components

1. User Layer

The EEE User represents the user whose access is being tested. The administrator is responsible for configuring and verifying the security settings.

2. Role Layer

The custom roles bb1, bb2, bb3, and bb4 represent different levels of permission.

- bb1 → Read
- bb2 → Create
- bb3 → Write
- bb4 → Delete

3. ACL Security Layer

The ACL layer evaluates the user's request and determines whether the requested operation should be permitted. Separate ACLs are configured for:

- Read
- Create
- Write
- Delete

4. Field Condition Layer

For the configured Read restriction, the Branch field is evaluated with the condition:

Branch = EEE

This provides the field-value-based component of the access-control mechanism.

5. Data Layer

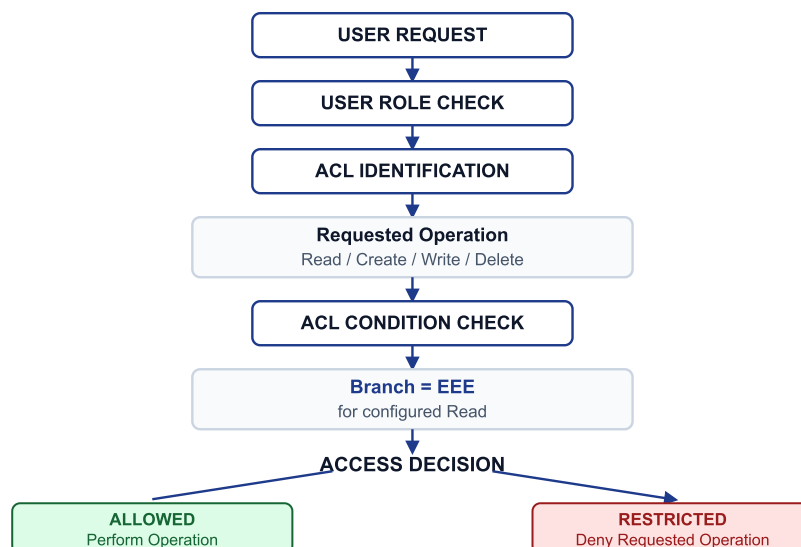
The Institution Details (u_institution_details) table contains the student records on which the ACLs operate.

The records include different Branch values so that the access restriction can be verified.

6. Access Decision Layer

After the relevant ACL conditions are evaluated, the system determines whether the requested operation is allowed or restricted.

Overall Solution Architecture Flow



Final Design Summary

The project design establishes a structured relationship between users, custom roles, ACLs, field conditions, and the Institution Details table. The design ensures that every requested operation passes through the appropriate access-control mechanism before the operation is permitted, thereby providing controlled and verifiable record-level security in ServiceNow.